Skills Summary

Measuring the Centre and Spread of a Data Set

Use this reference while completing your worksheet or when studying.

Skill 1 — Mode and median: the middle of the data

Order the data first. Every measure in this section is about position, so nothing works until the values are written smallest to largest.

Mode = the value that appears most often (it is a value from the list, not how many times it appeared). **Median** = the middle value once ordered. With an odd count that is the single middle one; with an even count, add the two middle values and halve.

Example: Find the median of $12, 15, 11, 20, 15, 13$.
Step 1. Six values is an even count, so there is no single middle one — order them, then average the two in the middle.
Step 2. Ordered: $11, 12, \mathbf{13}, \mathbf{15}, 15, 20$, and halfway between 13 and 15 is 14.
$\Rightarrow$ median = **14** (the mode here is 15, the only repeated value)

Try it: Find the median of $8, 5, 9, 6$.

check: median = **7**

Skill 2 — The mean: sharing the total out equally

Mean = (sum of all the values) $\div$ (how many values there are). It is the size every value *would* be if the total were shared out equally, so it need not equal any value in the list.

Because every value is added in, the mean feels each one. A single value far from the rest drags the mean towards it — the median cannot be moved that way.

Example: Find the mean of $6, 9, 4, 8, 3$.
Step 1. The mean shares the total out equally, so no ordering is needed — add everything, then divide by how many there are.
Step 2. $6 + 9 + 4 + 8 + 3$ comes to thirty, and there are five values, so share thirty among five.
$\Rightarrow$ mean = **6**

Try it: Find the mean of $12, 7, 10, 15$.

check: mean = **11**

Skill 3 — Range and interquartile range

Range = largest $-$ smallest. It uses only two values, so one unusual value decides it.

Quartiles: order the data, split it at the median (leave the median itself out of both halves when the count is odd). Q_1 is the median of the lower half, Q_3 the median of the upper half, and the **interquartile range** is $Q_3 - Q_1$ — the width of the middle half. Nothing at the far ends can change it.

Example: Find the range and the IQR of $2, 4, 5, 7, 8, 9, 20$.
Step 1. Two questions about spread: the range asks how far apart the extremes are, the IQR asks how wide the middle half is, so one uses the ends and the other uses the quartiles.
Step 2. Range: $20 - 2$. Median is 7; lower half $2, 4, 5$ gives $Q_1 = 4$; upper half $8, 9, 20$ gives $Q_3 = 9$; so IQR $= 9 - 4$.
$\Rightarrow$ range $=$ **18**, IQR $=$ **5** (the 20 decides the first and cannot touch the second)

Try it: Find the IQR of $1, 3, 4, 6, 7, 9, 10$.

check: IQR $=$ **6**

Skill 4 — Chance read off a data set

If one item is picked from a data set at random and every item is equally likely, then **P(event)** = (how many *items* match) ÷ (how many items there are).

Count items, never the different values on the axis: a value that four items share must be counted four times.

Example: One value is picked at random from the seven values in the example above. What is the probability it is greater than 7?
Step 1. Every value is equally likely, so this is just counting: how many of the seven values match, out of seven.
Step 2. Three of the seven values sit above that mark.
$\Rightarrow$ $P = \mathbf{\frac{3}{7}}$

Try it: From those same seven values, what is the probability of picking one less than 5?

check: $P = \mathbf{\frac{2}{7}}$